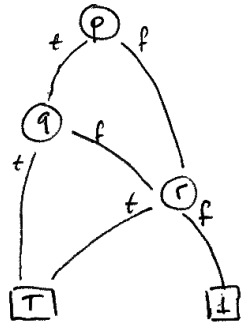
Nimi: _____

ITI 0402 Loogika, sügis 2026
Tunnikontroll 18.10.2026
Lahendused

1. Kirjuta järgmise taandatud binaarse otsustusdiagrammi pealt välja temaga samaväärne disjunktiivsel normaalkujul valem.



Vastus:

$(p \wedge q) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge r)$

2. Tõesta loomulikus tuletuses valem

$$p \wedge q \supset (\neg(q \vee r) \supset s)$$

Vastus:

$$\dfrac{\dfrac{\dfrac{\dfrac{+2}{\neg(q \vee r)} \quad \dfrac{\dfrac{\dfrac{+1}{p \wedge q}}{q} \wedge\mathcal{E}_R}{q \vee r} \vee\mathcal{I}_L}{\dfrac{\perp}{s} \perp\mathcal{E}} \neg\mathcal{E}}{\neg(q \vee r) \supset s} \supset\mathcal{I}, -2}{p \wedge q \supset (\neg(q \vee r) \supset s)} \supset\mathcal{I}, -1$$

3. Tõesta sekventsiarvutuses valem

$$(p \supset q \wedge r) \supset (s \wedge p \supset q)$$

Vastus:

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{s, p \to p, q}\ hyp. \quad \dfrac{\dfrac{}{q, r, s, p \to q}\ hyp.}{q \wedge r, s, p \to q} \wedge\mathcal{L}}{p \supset q \wedge r, s, p \to q} \supset\mathcal{L}}{p \supset q \wedge r, s \wedge p \to q} \wedge\mathcal{L}}{p \supset q \wedge r \to s \wedge p \supset q} \supset\mathcal{R}}{\to (p \supset q \wedge r) \supset (s \wedge p \supset q)} \supset\mathcal{R}$$